3.17 Cayley-Hamilton Theorem

Definition 3.17.1 (Characteristic Polynomial of a Matrix)
For a matrix $E \in \mathrm{Mat}_n(A)$ define the characteristic polynomial by

$$P_E(X) := \det(X \cdot I_n - E)$$

working in $\mathrm{Mat}_n(A[X])$. This is a monic polynomial in $A[X]$ and satisfies $P_E(X) = P_{E^t}(X)$ (3.8.20).

Definition 3.17.2 (Characteristic Polynomial of an endomorphism of a free module)
Let M be a finite free A-module. Let $\mathcal{B}$ be a basis for M, $\phi : M \to M$ an A-module endomorphism and $[\phi]$ the matrix representation of ϕ with respect to $\mathcal{B}$. Define the characteristic polynomial of ϕ by

$$P_\phi(X) := P_{[\phi]}(X)$$

This is independent of the basis $\mathcal{B}$ by (3.8.20).

Lemma 3.17.3
Suppose $M = \langle m_1, ..., m_n \rangle$ is a finitely generated A-module and $\mathfrak{a} \triangleleft A$ an ideal, then

$$\mathfrak{a}M = \mathfrak{a}m_1 + ... + \mathfrak{a}m_n$$

That is every $m \in \mathfrak{a}M$ may be written as

$$m = \sum_{i=1}^{n} a_i m_i \quad \text{for some } a_i \in \mathfrak{a}$$

Proof. By hypothesis

$$m = \sum_{j=1}^{n} b_j m'_j \quad \text{for some } m'_j \in M, b_j \in \mathfrak{a}$$

Furthermore by the finite-generation hypothesis for $j = 1...n$

$$m'_j = \sum_{i=1}^{n} c_{ji} m_i \quad \text{for some } c_{ji} \in A.$$

Therefore

$$m = \sum_{i=1}^{n} \left(\sum_{j=1}^{n} b_j c_{ji} \right) m_i$$

whence $a_i := \sum_{j=1}^{n} b_j c_{ji}$ lie in $\mathfrak{a}$ and give the required coefficients. ∎

Theorem 3.17.4 (Cayley-Hamilton)
Let M be a finitely generated A-module and $\phi \in \mathrm{End}_A(M)$. Then there exists a monic polynomial $P(X) \in A[X]$ such that $P(\phi) = 0$.

This may be strengthened in the following ways

a) If M is a finite free A-module then P may be taken to be the characteristic polynomial $P_\phi(X)$.
b) If $\phi(M) \subseteq \mathfrak{a}M$ for some ideal $\mathfrak{a} \triangleleft A$, then the non-leading coefficients of $P(X)$ may be chosen to be in $\mathfrak{a}$.

Proof. First since $\mathrm{End}_A(M)$ is an A-algebra there is a canonical evaluation morphism

$$\mathrm{ev}_\phi : A[X] \to \mathrm{End}_A(M)$$

and the meaning of $P(\phi)$ is simply $\mathrm{ev}_\phi(P)$.

Let $\{m_1, ..., m_n\}$ be a generating set, then by definition

$$\phi(m_i) = \sum_{j=1}^{n} E_{ij} m_j \quad i = 1...n$$

for some $E \in \mathrm{Mat}_n(A)$. Consider the matrix

$$B(X) = XI_n - E \in \mathrm{Mat}_n(A[X])$$

Then we may define $B(\phi) := B(X)^{\mathrm{ev}_\phi} \in \mathrm{Mat}_n(\mathrm{End}_A(M))$ pointwise, so given by

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$$B(\phi)_{ij} = \delta_{ij}\phi - E_{ij}1_M \quad i,j = 1...n$$

By definition

$$\sum_j B(\phi)_{ij}m_j = \phi(m_i) - \sum_j E_{ij}m_j = 0$$

Formally we have a group action

$$\begin{aligned} \mathrm{Mat}_n(\mathrm{End}_A(M)) \times M^n &\to M^n \\ F \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} &\to \begin{pmatrix} \sum_{j=1}^n F_{1j}(x_j) \\ \vdots \\ \sum_{j=1}^n F_{nj}(x_j) \end{pmatrix} \end{aligned}$$

such that $(EF)v = E(Fv)$ (check). And we have shown that

$$B(\phi)\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

Using (3.8.22), premultiply by the adjugate matrix to show that

$$\det(B(\phi))I_n\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

and $\det(B(\phi)) \in \mathrm{End}_A(M)$ annihilates $m_1, ..., m_n$ and therefore M.

Finally we claim that $P(X) := \det(B(X)) \in A[X]$ is a suitable monic polynomial. We see that

$$P(\phi) := \mathrm{ev}_\phi(\det(B(X))) = \det(B(X)^{\mathrm{ev}_\phi}) = \det(B(\phi)) = 0$$

When M is a finite free A-module then we may choose $\{m_1, ..., m_n\}$ to be a basis, and then the matrix E equals $[\phi]^T$ as required.

Finally when $\phi(M) \subseteq \mathfrak{a}M$ then (3.17.3) shows we may choose the coefficients E_{ij} to be in $\mathfrak{a}$. It's clear that $P(X)$ then has non-leading coefficients in $\mathfrak{a}$. ∎

3.17.1 The $A[X]$-module associated to an endomorphism

Proposition 3.17.5
Let $\phi : M \to M$ be an A-module endomorphism. Then there is a well-defined $A[X]$-module structure defined by

$$\begin{aligned} A[X] \times M &\to M \\ (P, m) &\to P(\phi)(m) \end{aligned}$$

where $P = \sum_{i=0}^n a_i X^i$. We denote this by M_ϕ.

With this notation then ϕ is itself an $A[X]$-module endomorphism of M_ϕ.

Proof. Let $P, Q \in A[X]$ be two polynomials. Then we need to show that

$$(PQ)(\phi) = P(\phi) \circ Q(\phi)$$

which amounts to saying that $A[X] \to \mathrm{End}_A(M)$ is an A-algebra homomorphism.

The final statement follows because X commutes with any polynomial in $A[X]$. More explicitly

$$\phi(P \cdot m) = \phi(P(\phi)(m)) = (XP)(\phi)(m) = (PX)(\phi)(m) = P(\phi)(\phi(m)) = P \cdot (\phi(m))$$

∎

Proposition 3.17.6
Let M be an A-module and $\phi \in \mathrm{End}_A(M)$. Then is an $A[X]$-linear map

$$\begin{aligned} \tilde{\phi} : A[X] \otimes_A M &\to M_\phi \\ P \otimes m &\to P(\phi)(m) \end{aligned}$$

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and similarly

$$1 \otimes \phi : A[X] \otimes_A M \to A[X] \otimes_A M$$
$$P \otimes m \to P \otimes (\phi(m))$$

Proof. By the universal property $\tilde{\phi}$ exists and is A-linear. To show it is $A[X]$-linear observe that

$$\tilde{\phi}(Q(P \otimes m)) = \tilde{\phi}((QP) \otimes m) = (QP)(\phi)(m) = Q(\phi)(P(\phi)(m)) = Q \cdot P(\phi)(m) = Q \cdot \tilde{\phi}(P \otimes m)$$

and so the result follows by linearity.

The second exists by (3.4.17). ∎

Proposition 3.17.7
Let M be a finite free A-module and $\phi \in \mathrm{End}_A(M)$. Then the characteristic polynomial $P_\phi(X)$ equals the determinant of the $A[X]$-linear map

$$(X - 1 \otimes \phi) : A[X] \otimes_A M \to A[X] \otimes_A M$$

Furthermore

$$P_\phi(X) = X^n + \sum_{j=1}^{n} (-1)^j \operatorname{Tr}(L_a^j(\phi)) X^{n-j}$$

and in particular

$$P_\phi(X) = X^n - \operatorname{Tr}(\phi) X^{n-1} + ... + (-1)^n \det(\phi)$$

Proof. Let $\mathcal{B} := \{v_1, ..., v_n\}$ be a basis for M, then by (3.4.24) $\{1 \otimes v_1, ..., 1 \otimes v_n\}$ is an $A[X]$-basis for $A[X] \otimes_A M$. Let $E = [\phi]$ be the matrix associated to the basis $\mathcal{B}$ then by definition

$$\phi(v_i) = \sum_{j=1}^{n} E_{ji} v_j$$

and

$$(X - 1 \otimes \phi)(1 \otimes v_i) = X \otimes v_i - 1 \otimes \phi(v_i) = X(1 \otimes v_i) - \sum_{j=1}^{n} E_{ji}(1 \otimes v_j)$$

This shows that

$$[X - 1 \otimes \phi]_{ij} = X\delta_{ij} - E_{ij}$$

and so $P_\phi(X) = \det(XI_n - E) = \det(X - 1 \otimes \phi)$ as required.

Then by (3.8.32) we have

$$P_\phi(X) = X^n + \sum_{j=1}^{n} (-1)^j \operatorname{Tr}(L_a^j(1 \otimes \phi)) X^{n-j}$$

The matrix of $1 \otimes \phi$ coincides with that of ϕ and so we deduce the second relationship. The final statement follows from (...). ∎

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